EXACT EXCLUSION OF 43,046,721 CROSS-BLOCK VOLTAGE GRAPHS FOR THE ERDŐS–GYÁRFÁS 2^k-CYCLE CONJECTURE

ARGUS OPEN RESEARCH PROJECT

Abstract. The Erdős–Gyárfás conjecture asserts that every graph of minimum degree at least 3 contains a simple cycle of length a power of 2. We prove that in the cross-block spoke voltage family over $\mathbb{Z}_{32}$, every one of the $3^{16} = 43{,}046{,}721$ voltage assignments contains a simple cycle of length 8, 16, or 32, covered by nine explicit cycle witnesses without symmetry reduction.

1. Main Family Exclusion

Theorem 1.1. *Across the* $3^{16} = 43{,}046{,}721$ *cubic voltage lifts in the cross-block spoke family over* $\mathbb{Z}_{32}$*, nine explicit closed voltage walk templates lift to simple cycles of length* 8*,* 16*, or* 32 *that partition the entire assignment space, proving that no graph in this family is a counterexample to the Erdős–Gyárfás conjecture.*

Date: September 5, 2026.
2020 *Mathematics Subject Classification.* Primary 05C38; Secondary 05C25.